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Fuzzy Multiple Linear Least Squares Regression Analysis in the Hilbert Space of A-Linearly Correlated Fuzzy Numbers

***Abstract*—This paper presents a fuzzy least squares method for multiple linear regression models formulated in the space of A-linearly correlated fuzzy numbers, denoted by $\mathbb{R}_{\mathcal{F}(A)}$. The algebraic operations in $\mathbb{R}_{\mathcal{F}(A)}$ are defined through an isometric isomorphism with the complex field $\mathbb{C}$, guaranteeing that $\mathbb{R}_{\mathcal{F}(A)}$ is a field and that $\mathbb{R}^p_{\mathcal{F}(A)}$ is a Hilbert space over $\mathbb{R}_{\mathcal{F}(A)}$. As a consequence, all the tools of differential and integral calculus for functions with fuzzy input and output become available, as well as the full framework of orthogonal projection theory in this class of fuzzy numbers. Based on these fundamentals, we propose a least squares method for fuzzy multiple linear regression models, where the coefficients can be either real or fuzzy numbers. Numerical results show that the proposed $\mathbb{R}_{\mathcal{F}(A)}$-based formulation yields lower Root Mean Squared Error (RMSE) values in comparison with existing approaches, lower Mean Absolute Error (MAE) values under the $\mathbb{R}_{\mathcal{F}(A)}$ norm, and higher agreement with the observed fuzzy outputs, as measured by similarity.**

***Index Terms*—Fuzzy multiple linear regression, Least squares estimation, A-linearly correlated fuzzy numbers, Root mean squared error, Mean absolute error, Similarity measures.**

I. Introduction

The treatment of uncertain data through fuzzy numbers based on least squares method has been investigated since the 1980s, with foundational contributions that established the theoretical and computational basis for this approach [1]–[5]. In many practical problems, observations exhibit imprecision or uncertainty, making it natural to model them using fuzzy numbers. In such contexts, classical estimation procedures must be extended to a fuzzy framework. In particular, least squares methods should be reformulated in a way that remains both mathematically coherent and computationally feasible.

Both traditional and contemporary studies have explored variations of the fuzzy least squares problem, considering different assumptions regarding the shape of fuzzy numbers (triangular, trapezoidal, or sigmoidal), the admissible algebraic operations and the metrics used to evaluate residuals and goodness of fit [4]–[14]. Several authors have investigated models in which both input and output variables are fuzzy [2], [5], [6], [10], [12], [13]. Additionally, some represent the coefficients of the model as fuzzy numbers [2], [4], [6].

When experimental data are represented by pairs of fuzzy numbers (X_i, Y_i), a central question concerns how to define a measure of fit between observations and predictions, as well as the algebraic operations necessary to minimize residuals. A general formulation postulates linear models of the type

$$Y = BX + C, \tag{1}$$

where B, X, and C are fuzzy numbers. In frameworks constructed via Zadeh's extension principle [15], the product BX may generate membership functions whose geometry differs from the original shapes (such as triangular or trapezoidal), which has motivated criticism regarding the direct applicability of model (1) to fuzzy pairs (X_i, Y_i) (see, for example, [2], [3], [5], [8]).

Moreover, forcing non-negative regression coefficients in fuzzy regression (aimed at preserving valid spreads) can lead to both interpretative and computational difficulties, as highlighted by Diamond [5]. Such issues are often linked to inconsistencies arising from the application of Zadeh's extension principle [5]. In contrast, adopting alternative operational frameworks, such as those based on the generalized Hukuhara difference (gH) [5] and interactive operations [16], leads to distinct algebraic behaviors.

In particular, interactive operations can be defined on the class of A-linearly correlated fuzzy numbers $\mathbb{R}_{\mathcal{F}(A)} = \{r + qA \mid (r, q) \in \mathbb{R}^2\}$. It has been shown that $\mathbb{R}_{\mathcal{F}(A)}$ is a complete vector space (a Banach space) [17] and that there exists an isometric identification between $\mathbb{R}_{\mathcal{F}(A)}$ and the field of complex numbers $\mathbb{C}$, when A is an asymmetric fuzzy number, enabling the use of classical analytical and numerical tools in this space [14], [18], [19]. Owing to this isomorphism, the least squares problem for fuzzy linear regression models in $\mathbb{R}_{\mathcal{F}(A)}$ can be treated using the norm induced by the inner product inherited from $\mathbb{C}$ [14], thereby restoring the formalism of orthogonal projections and least squares estimators in the fuzzy context.

Various approaches have sought to improve least squares methods in fuzzy environments to handle the inherent nature and uncertainty of the data. Diamond et al. [5] proposed an extension of the fuzzy linear regression model by incorporating the generalized Hukuhara difference and a L^2-type metric, obtaining fuzzy least squares solutions in partially linear spaces. Yoon et al. [10] advanced this line of research by developing fuzzy least squares estimators based on new fuzzy operations, defining triangular matrices (based on triangular fuzzy numbers) and appropriate metrics to handle fuzzy input-output data. More recently, Li et al. [13] proposed a fuzzy multiple linear regression model via fuzzy least squares based on LR-fuzzy numbers, using two distance measures and introducing error and similarity indices to evaluate goodness of fit. Although these approaches represent significant advances in the formulation and computational treatment of the problem, they all share a common feature: only the intercept is considered a fuzzy number, while the remaining coefficients remain real.

This article extends the fuzzy least squares formulation introduced in $\mathbb{R}_{\mathcal{F}(A)}$ [14] to the multivariate linear case, in which both the data and the coefficients are fuzzy. We derive the corresponding normal equations, whose solution yields the regression coefficients estimated by the fuzzy least squares

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method. This framework parallels the classical approach to parameter estimation in regression analysis, while incorporating the fuzziness inherent in the data and model parameters within a unified Hilbert space structure.

From isomorphism between $\mathbb{C}$ and $\mathbb{R}_{\mathcal{F}(A)}$ (see [18]), the principal contributions of the present study are

(1) The space $\mathbb{R}_{\mathcal{F}(A)}$ is a field, and $\mathbb{R}^p_{\mathcal{F}(A)}$ is a p-dimensional Hilbert space over the field $\mathbb{R}_{\mathcal{F}(A)}$ (equivalently, a $2p$-dimensional Hilbert space over the field $\mathbb{R}$);
(2) The estimation from fuzzy least squares method is obtained by orthogonal projection (best approximation) of output vector onto the subspace generated by the columns of the input matrix;
(3) In the case where the coefficients are real numbers, the minimization can be carried out using differential-calculus methods applied to the parameters;
(4) The method accommodates both symmetric and asymmetric datasets and allows for comparison with recent approaches in fuzzy least squares methods and similarity measures.

The implementation is straightforward via standard matrix operations on the normal equations.

The structure of the paper is as follows: Section II introduces the preliminary concepts and fundamental algebraic properties of $\mathbb{R}_{\mathcal{F}(A)}$. Section III compares the proposed methodology with the conventional approach based on LR-fuzzy numbers. Section IV formulates the fuzzy least squares method with real coefficients and presents its matrix representation. The formulation is then extended to fuzzy coefficients, together with a discussion of the associated algebraic and analytical implications. Subsection V-A presents numerical applications involving real coefficients and comparisons with existing methods, followed by numerical applications with fuzzy coefficients in Subsection V-B. Finally, Section VI concludes the paper.

II. Preliminaries

A fuzzy subset of $\mathbb{R}$ is defined as a map, called a membership function, from $\mathbb{R}$ to $[0,1]$. Thus, we identify a fuzzy subset A with its membership function $\mu_A(x)$. For any $\alpha \in (0,1]$, the crisp set $[A]_\alpha = \{x \in \mathbb{R} \mid \mu_A(x) \geqslant \alpha\}$, is called the α-level set of A. The set $[A]_0 = \overline{\{x \in \mathbb{R} \mid \mu_A(x) > 0\}}$, is the closure of the support of A. A fuzzy set A is called a fuzzy number if $[A]_\alpha$ is non-empty, bounded closed interval of $\mathbb{R}$ for all $\alpha \in [0,1]$ [20]. The collection of all fuzzy numbers will be denoted by $\mathbb{R}_{\mathcal{F}}$. A fuzzy number A is said to be *asymmetric* (or non-symmetric) if, for all $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that $\mu_A(x-y) \neq \mu_A(x+y)$. The set of asymmetric fuzzy numbers forms an open and dense subset of the space of all fuzzy numbers [21]. From a practical standpoint, this subset is large enough to approximate any fuzzy number, and asymmetry is preserved under small perturbations. In this case, the asymmetric fuzzy set is referred to as *quasi-symmetric*.

For each $A \in \mathbb{R}_{\mathcal{F}}$, we define the operator $\Phi_A : \mathbb{C} \to \mathbb{R}_{\mathcal{F}}$ that associates each complex number $(r+iq) \in \mathbb{C}$ with the fuzzy number

$$\Phi_A(r+iq) = r + qA, \quad \forall\ r, q \in \mathbb{R}.$$

The range of this operator is denoted by $\mathbb{R}_{\mathcal{F}(A)}$. For notational convenience, we use the symbol Φ instead of Φ_A. Note that $\mathbb{R} \subset \mathbb{R}_{\mathcal{F}(A)}$, since every real number r can be identified with the fuzzy number $\Phi(r+i0) \in \mathbb{R}_{\mathcal{F}(A)}$. By analogy with complex numbers, we use $r = \mathrm{Re}(r+qA)$ and $q = \mathrm{Fu}(r+qA)$, that is, the real and fuzzy parts of the fuzzy number $B = r + qA$.

Theorem 1. *[17], [18] Given $A \in \mathbb{R}_{\mathcal{F}}$, consider the operator $\Phi : \mathbb{C} \to \mathbb{R}_{\mathcal{F}(A)}$ given by*

$$\Phi(r+iq) = r + qA, \tag{2}$$

for all $z = (r+iq) \in \mathbb{C}$. The operator Φ is a bijection, if and only if, A is asymmetric.

Before proceeding, it is important to note that the construction of the space $\mathbb{R}_{\mathcal{F}(A)}$ does not depend on the specific asymmetric fuzzy number chosen in the definition of Φ. In other words, if B is non-real element in $\mathbb{R}_{\mathcal{F}(A)}$, then $\{1, A\}$ and $\{1, B\}$ generate the same space. This observation is made precise in the following remark.

Remark 1. *For every $B \in \mathbb{R}_{\mathcal{F}(A)} \backslash \mathbb{R}$, the range of Φ_B does not depend on the choice of B, since $\mathbb{R}_{\mathcal{F}(B)} = \mathbb{R}_{\mathcal{F}(A)}$ [17].*

Theorem 2. *[18] Let A be an asymmetric fuzzy number. The set $\mathbb{R}_{\mathcal{F}(A)}$ is a field isomorphic to $\mathbb{C}$ where the arithmetic operations are given by*

$$B \circledast C = \Phi(\Phi^{-1}(B) * \Phi^{-1}(C)), \tag{3}$$

for every $B, C \in \mathbb{R}_{\mathcal{F}(A)}$, where $*$ stands one of the four arithmetic operations of $\mathbb{C}$, i.e., $* \in \{+, -, \cdot, \div\}$. In addition, $\mathbb{R}_{\mathcal{F}(A)}$ is isometric to $\mathbb{C}$ with the norm

$$\|B\|_\Phi = |\Phi^{-1}(B)| = \sqrt{r^2 + q^2} = \sqrt{(Re(B))^2 + (Fu(B))^2}. \tag{4}$$

Remark 2. *The $\odot$ multiplication operator naturally extends the usual scalar multiplication in $\mathbb{R}_{\mathcal{F}(A)}$. Indeed, for any real scalar $\lambda \in \mathbb{R} \subset \mathbb{R}_{\mathcal{F}(A)}$ and any fuzzy number $B = r + qA \in \mathbb{R}_{\mathcal{F}(A)}$, one has*

$$\lambda \odot B = (\lambda + 0A) \odot (r + qA) = \lambda r + \lambda q\, A = \lambda(r + qA) = \lambda B.$$

In the literature [15], [22], basic operations on fuzzy numbers are defined through Zadeh's extension principle. However, the main advantage of working within $\mathbb{R}_{\mathcal{F}(A)}$ is that, under the weak asymmetry condition of A and through the isomorphism Φ, one can employ algebraic operations analogous to those in the complex field $\mathbb{C}$. Explicitly, by Theorem 2, with $B = r + qA$ and $C = s + pA$, each arithmetic operation in $\mathbb{R}_{\mathcal{F}(A)}$ can be written as

$$B \oplus C = (r+s) + (q+p)A, \tag{5}$$
$$B \ominus C = (r-s) + (q-p)A, \tag{6}$$
$$B \odot C = (rs - qp) + (sq + rp)A, \tag{7}$$
$$B \oslash C = \frac{rs + qp}{s^2 + p^2} + \frac{qs - rp}{s^2 + p^2}A, \tag{8}$$

provided that $C \neq 0$ in Eq. (8). Despite the analogy between the arithmetic of the field of complex numbers and fuzzy numbers in $\mathbb{R}_{\mathcal{F}(A)}$, unlike in the complex case, the expression $r + qA$ represents the sum of the fuzzy numbers r and qA, yielding another fuzzy number B without the

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need to separate real and fuzzy components. For example, $B = 2 + 3(-1;0;3)_T = (-1;2;11)_T$, where $B = (a;b;c)_T$ represents the triangular fuzzy number with $[B]_0 = [a,c]$ and $[B]_1 = \{b\}$. In this representation, the scalar r determines the central value of the fuzzy number B, while q scales the dispersion induced by the generating fuzzy number A.

The conjugate of $B = r + qA \in \mathbb{R}_{\mathcal{F}(A)}$ is defined by $\overline{B} = r - qA$, and will be used in the following definition of the inner product on $\mathbb{R}^p_{\mathcal{F}(A)}$. These concepts provide a foundation for the fuzzy least squares criterion adopted in fuzzy linear regression.

Definition 1 (Inner product)**.** *The inner product on $\mathbb{R}^p_{\mathcal{F}(A)}$ is defined by*

$$\langle z, w\rangle = \bigoplus_{i=1}^{p} z_i \odot \overline{w}_i, \qquad \forall\, z, w \in \mathbb{R}^p_{\mathcal{F}(A)}, \tag{9}$$

where $\overline{w}_i$ denotes the conjugate of $w_i \in \mathbb{R}_{\mathcal{F}(A)}$.

Now, as in the complex case [23], $\mathbb{R}^p_{\mathcal{F}(A)}$ endowed with the inner product $\langle\cdot,\cdot\rangle$ is a Hilbert space [14], [18]. In fact, from the isomorphism Φ, the operator $\|z\|_\Phi = \sqrt{\langle z, z\rangle}$ for $z \in \mathbb{R}^p_{\mathcal{F}(A)}$ is a norm on $\mathbb{R}^p_{\mathcal{F}(A)}$.

Due to the importance of the LR class of fuzzy numbers in fuzzy linear regression, we will discuss it briefly below.

III. Comments about LR-fuzzy numbers and $\mathbb{R}_{\mathcal{F}(A)}$

For comparison with traditional approaches in numerical applications, we briefly recall the class of LR-fuzzy numbers. These are fuzzy numbers whose membership functions are defined by [7], [13]

$$\mu_A(x) = \begin{cases} L\Big(\dfrac{m-x}{l}\Big), & \text{if } x \leqslant m, \\ R\Big(\dfrac{x-m}{u}\Big), & \text{if } x > m, \end{cases} \tag{10}$$

where m is the modal point, $l, u \geqslant 0$ are the left and right spreads, respectively. The pre-defined functions $L(\cdot)$ and $R(\cdot)$, known as the left and right shape functions, satisfy the following properties

(1) $L(\cdot)$ and $R(\cdot)$ are decreasing from $\mathbb{R}^+ \to [0,1]$;
(2) $L(0) = R(0) = 1$ and $L(x), R(x) > 0$ for $x > 0$;
(3) $L(1) = R(1) = 0$ and $L(x), R(x) > 0$ for $x < 1$;
(4) $L^{-1}(\alpha) = \sup\{x \in \mathbb{R} : L(x) \geqslant \alpha\}$ and $R^{-1}(\alpha) = \sup\{x \in \mathbb{R} : R(x) \geqslant \alpha\}$.

We represent this class of fuzzy numbers by $A = (m;l;u)_{LR}$.

As a particular case, when the shape functions satisfy $L(x) = R(x) = \max(0, 1-|x|)$, the fuzzy number is triangular and is denoted by $A = (m;l;u)_T$. The Gaussian form is defined by the membership function

$$\mu_A(x) = \begin{cases} \exp\bigg(-\Big(\dfrac{x-m}{l}\Big)^2\bigg), & m-l \leqslant x \leqslant m, \\ \exp\bigg(-\Big(\dfrac{x-m}{u}\Big)^2\bigg), & m < x \leqslant m+u, \\ 0, & \text{otherwise}, \end{cases} \tag{11}$$

where l and u denote the left and right spreads. If $l \approx u$, the fuzzy number is *quasi-normal* and is written $A = (m;l;u)_N$, whereas in the symmetric case $l = u = \sigma$ it reduces to the *normal* form $A = (m;\sigma)_N$.

Remark 3. *Throughout this work, a fuzzy number will be denoted either in the compact form $(m;l;u)_{LR}$ or equivalently as $(m-l;m;m+u)_{LR}$. The specific notation adopted will be explicitly stated whenever necessary.*

It is worth noting that, while the $\mathbb{R}_{\mathcal{F}(A)}$ fixes the asymmetric fuzzy number A, the LR framework fixes the shape functions L and R in advance. However, the algebraic and topological structures of these classes differ: while $\mathbb{R}_{\mathcal{F}(A)}$ forms a field with arithmetic operations induced by Φ (Theorem 2), the LR class forms a metric space where arithmetic is given by Zadeh's extension principle [5], [13].

In the fuzzy least squares approach, the fuzzy number A in $\mathbb{R}_{\mathcal{F}(A)}$ may adopt various asymmetric shapes (triangular, trapezoidal, sigmoidal, etc.), whereas some classical formulations restrict the fuzzy shape to the triangular case (see, e.g., [10]). A key distinction lies in the metric structure: while classical LR-based methods rely on externally defined distance measures between fuzzy numbers, in $\mathbb{R}_{\mathcal{F}(A)}$ the norm $\|\cdot\|_\Phi$ is intrinsic to the space through the isometry Φ from Theorem 2, ensuring full algebraic–metric consistency throughout the estimation procedure.

To evaluate the performance of the adopted model in numerical applications, it is customary to use the Mean Absolute Error (MAE) and the Root Mean Squared Error (RMSE) metrics

$$\mathrm{MAE}_d = \frac{1}{p}\sum_{i=1}^{p} d(y_i, \hat{y}_i), \quad \mathrm{RMSE}_d = \sqrt{\frac{1}{p}\sum_{i=1}^{p} d(y_i, \hat{y}_i)^2}, \tag{12}$$

where p is the number of data points, y_i represents the observed values, $\hat{y}_i$ denotes the predicted values and $d(y_i, \hat{y}_i)$ gives the distance between y_i and $\hat{y}_i$ based on the metric d of the space of interest. In our framework on the $\mathbb{R}_{\mathcal{F}(A)}$ space, Equations (12) become

$$\mathrm{MAE}_{\|\cdot\|_\Phi} = \frac{1}{p}\sum_{i=1}^{p} \|y_i \ominus \hat{y}_i\|_\Phi, \quad \mathrm{RMSE}_{\|\cdot\|_\Phi} = \sqrt{\frac{1}{p}\sum_{i=1}^{p} \|y_i \ominus \hat{y}_i\|^2_\Phi}. \tag{13}$$

Additionally, it is also common to employ similarity measures to quantify how close the predicted fuzzy outputs are to the observed fuzzy data. Here, we adopt the similarity measures introduced in [13], which provide a complementary perspective on model performance by evaluating the degree of overlap between fuzzy sets.

IV. Fuzzy Multiple Linear Least Squares in $\mathbb{R}_{\mathcal{F}(A)}$

Consider an asymmetric fuzzy dataset

$$\mathcal{D} = \{(x_i, y_i) \in \mathbb{R}^n_{\mathcal{F}(A)} \times \mathbb{R}_{\mathcal{F}(A)} \mid i = 1, \ldots, p\},$$

where each $x_i = (x_{i1}, \ldots, x_{in})^\mathsf{T} \in \mathbb{R}^n_{\mathcal{F}(A)}$ is a vector of fuzzy inputs and $y_i \in \mathbb{R}_{\mathcal{F}(A)}$ is the corresponding fuzzy output.

We model the relationship between the fuzzy inputs and fuzzy output using a multiple linear regression model with n independent variables and $n+1$ coefficients

$$y_i = \underbrace{\alpha_0 \oplus \alpha_1 \odot x_{i1} \oplus \cdots \oplus \alpha_n \odot x_{in}}_{\hat{y}_i} \oplus \epsilon_i, \tag{14}$$

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where $\alpha_0, \ldots, \alpha_n$ are the coefficients, which can be either all real or all fuzzy and $\epsilon_i \in \mathbb{R}_{\mathcal{F}(A)}$ represents the fuzzy residual, for $i = 1, \ldots, p$.

In matrix form, Equation (14) can be written as

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix} = \left(\begin{bmatrix} 1 & x_{11} & \ldots & x_{1n} \\ 1 & x_{21} & \ldots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{p1} & \ldots & x_{pn} \end{bmatrix} \odot \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \right) \oplus \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_p \end{bmatrix}. \quad (15)$$

or more compactly

$$\mathbf{y} = (\mathbf{X} \odot \boldsymbol{\alpha}) \oplus \boldsymbol{\epsilon}, \quad (16)$$

with $\mathbf{y} \in \mathbb{R}^p_{\mathcal{F}(A)}$ being the vector of fuzzy outputs, $\mathbf{X} = [x_{ij}] \in \mathbb{R}^{p\times(n+1)}_{\mathcal{F}(A)}$ the fuzzy design matrix, following the usual convention that $x_{i0} = 1$ for all i, and $\boldsymbol{\alpha} \in \mathbb{R}^{n+1}_{\mathcal{F}(A)}$ being the coefficient vector.

The fuzzy least squares method consists in determining $\boldsymbol{\alpha} = (\alpha_0, \ldots, \alpha_n)^\mathsf{T}$ so that the norm of the fuzzy residual ($\|\mathbf{y} \ominus \mathbf{X} \odot \boldsymbol{\alpha}\|^2_\Phi$) is minimal, that is, to solve the minimization problem

$$\min_{\boldsymbol{\alpha} \in \mathbb{R}^{n+1}_{\mathcal{F}(A)}} \frac{1}{2} \|\mathbf{y} \ominus (\mathbf{X} \odot \boldsymbol{\alpha})\|^2_\Phi = \min_{\boldsymbol{\alpha} \in \mathbb{R}^{n+1}_{\mathcal{F}(A)}} F(\boldsymbol{\alpha}). \quad (17)$$

Depending on the nature of the coefficients α_j, for $j = 0, \ldots, n$, we distinguish two estimation scenarios. In the first case, the coefficients are real numbers, which allows us to formulate a standard least squares problem using the real and fuzzy components of the data. This scenario is addressed in the next subsection.

A. Fuzzy Multiple Linear Least Squares with Real Coefficients in $\mathbb{R}_{\mathcal{F}(A)}$

In this case, the coefficient vector $\boldsymbol{\alpha} = (\alpha_0, \ldots, \alpha_n)^\mathsf{T} \in \mathbb{R}^{n+1} \subset \mathbb{R}^{n+1}_{\mathcal{F}(A)}$ consists of real components, and therefore the operation $\mathbf{X} \odot \boldsymbol{\alpha} = \mathbf{X}\alpha$ (see Remark 2) holds, so that the fuzzy multiple linear regression problem reduces to estimating these real parameters while still accounting for the fuzziness present in the dataset. Thus, Problem (17) becomes

$$\min_{\boldsymbol{\alpha} \in \mathbb{R}^{n+1}} \frac{1}{2} \|\mathbf{y} \ominus \mathbf{X}\boldsymbol{\alpha}\|^2_\Phi = \min_{\boldsymbol{\alpha} \in \mathbb{R}^{n+1}} F(\boldsymbol{\alpha}), \quad (18)$$

where the objective function $F : \mathbb{R}^{n+1} \to \mathbb{R}$ is defined on the real domain.

We write each entry of the design matrix $\mathbf{X}$ and the fuzzy response $\mathbf{y}$ as

$$x_{ij} = x^{(r)}_{ij} + x^{(f)}_{ij} A, \quad y_i = y^{(r)}_i + y^{(f)}_i A. \quad (19)$$

To make the problem more tractable [18], we separate the real and fuzzy components of both inputs and outputs. Denoting the real and fuzzy response of $\mathbf{y}$ by

$$\mathbf{y}^{(r)} = (y^{(r)}_1, \ldots, y^{(r)}_p)^\mathsf{T}, \qquad \mathbf{y}^{(f)} = (y^{(f)}_1, \ldots, y^{(f)}_p)^\mathsf{T}, \quad (20)$$

and defining the real and fuzzy design matrices of $\mathbf{X}$ as

$$\mathbf{X}^{(r)} = \begin{bmatrix} 1 & x^{(r)}_{11} & \cdots & x^{(r)}_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x^{(r)}_{p1} & \cdots & x^{(r)}_{pn} \end{bmatrix}, \mathbf{X}^{(f)} = \begin{bmatrix} 0 & x^{(f)}_{11} & \cdots & x^{(f)}_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & x^{(f)}_{p1} & \cdots & x^{(f)}_{pn} \end{bmatrix}, \quad (21)$$

the objective function $F(\boldsymbol{\alpha})$ can be rewritten component-wise by

$$\begin{aligned} F(\boldsymbol{\alpha}) &= \frac{1}{2}\|\mathbf{y}^{(r)} - \mathbf{X}^{(r)}\boldsymbol{\alpha}\|^2_2 + \frac{1}{2}\|\mathbf{y}^{(f)} - \mathbf{X}^{(f)}\boldsymbol{\alpha}\|^2_2 \\ &= \frac{1}{2}\big(\mathbf{y}^{(r)} - \mathbf{X}^{(r)}\boldsymbol{\alpha}\big)^\mathsf{T}\big(\mathbf{y}^{(r)} - \mathbf{X}^{(r)}\boldsymbol{\alpha}\big) \quad (22) \\ &+ \frac{1}{2}\big(\mathbf{y}^{(f)} - \mathbf{X}^{(f)}\boldsymbol{\alpha}\big)^\mathsf{T}\big(\mathbf{y}^{(f)} - \mathbf{X}^{(f)}\boldsymbol{\alpha}\big). \end{aligned}$$

To obtain the minimizer, we differentiate $F(\boldsymbol{\alpha})$ with respect to $\boldsymbol{\alpha}$ and set the gradient to zero. Then

$$\begin{aligned} \nabla_{\boldsymbol{\alpha}} F(\boldsymbol{\alpha}) = &-(\mathbf{X}^{(r)})^\mathsf{T}\big(\mathbf{y}^{(r)} - \mathbf{X}^{(r)}\boldsymbol{\alpha}\big) \\ &-(\mathbf{X}^{(f)})^\mathsf{T}\big(\mathbf{y}^{(f)} - \mathbf{X}^{(f)}\boldsymbol{\alpha}\big) = \mathbf{0}. \end{aligned}$$

Rearranging terms leads to the normal equations

$$\big((\mathbf{X}^{(r)})^\mathsf{T}\mathbf{X}^{(r)} + (\mathbf{X}^{(f)})^\mathsf{T}\mathbf{X}^{(f)}\big)\,\boldsymbol{\alpha} = (\mathbf{X}^{(r)})^\mathsf{T}\mathbf{y}^{(r)} + (\mathbf{X}^{(f)})^\mathsf{T}\mathbf{y}^{(f)}. \quad (23)$$

If we take the stacked notation

$$\tilde{\mathbf{y}} = (y^{(r)}_1, y^{(f)}_1, \ldots, y^{(r)}_p, y^{(f)}_p)^\mathsf{T} \in \mathbb{R}^{2p}, \quad (24)$$

and define the augmented design matrix

$$\mathbf{G} = \begin{bmatrix} 1 & x^{(r)}_{11} & x^{(r)}_{12} & \cdots & x^{(r)}_{1n} \\ 0 & x^{(f)}_{11} & x^{(f)}_{12} & \cdots & x^{(f)}_{1n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x^{(r)}_{p1} & x^{(r)}_{p2} & \cdots & x^{(r)}_{pn} \\ 0 & x^{(f)}_{p1} & x^{(f)}_{p2} & \cdots & x^{(f)}_{pn} \end{bmatrix} \in \mathbb{R}^{2p\times(n+1)}, \quad (25)$$

the normal equations (23) can be written compactly as

$$\mathbf{G}^\mathsf{T}\mathbf{G}\,\boldsymbol{\alpha} = \mathbf{G}^\mathsf{T}\tilde{\mathbf{y}}. \quad (26)$$

Geometrically, the minimization problem in (18) is equivalent to projecting the vector $\tilde{\mathbf{y}} \in \mathbb{R}^{2p}$ onto the column space of the augmented design matrix $\mathbf{G}$. Assuming $\mathbf{G}^\mathsf{T}\mathbf{G}$ is nonsingular, the fuzzy least squares estimator for linear regression is

$$\hat{\boldsymbol{\alpha}} = (\mathbf{G}^\mathsf{T}\mathbf{G})^{-1}\mathbf{G}^\mathsf{T}\tilde{\mathbf{y}}. \quad (27)$$

In the case where $\mathbf{G}$ is rank-deficient, the Moore-Penrose pseudo-inverse provides the generalized solution.

Hence, estimating the real coefficients of the fuzzy multiple linear regression model reduces to solving the standard least squares problem constructed from the real and fuzzy components of the dataset.

Example 1. *Let A be an asymmetric fuzzy number. Consider a fuzzy dataset composed of asymmetric fuzzy numbers*

$$\mathcal{D} = \{(x_{i1}, x_{i2}, y_i) \in \mathbb{R}^2_{\mathcal{F}(A)} \times \mathbb{R}_{\mathcal{F}(A)} \mid i = 1, 2, 3, 4\},$$

with two fuzzy input variables and one fuzzy output given by

$$\begin{aligned} &x_{11} = 1.0 + 0.3A, && x_{12} = 2.0 + 0.8A, && y_1 = 2.8 + 0.4A, \\ &x_{21} = 2.0 + 1.2A, && x_{22} = 1.5 + 0.1A, && y_2 = 3.7 + 1.3A, \\ &x_{31} = 3.0 + 0.5A, && x_{32} = 2.5 + 1.9A, && y_3 = 5.0 + 0.8A, \\ &x_{41} = 4.0 + 0.9A, && x_{42} = 3.0 + 0.6A, && y_4 = 6.1 + 2.7A, \end{aligned}$$

Accordingly, the fuzzy multiple linear model is assumed in the form

$$y_i = \alpha_0 \oplus \alpha_1 x_{i1} \oplus \alpha_2 x_{i2}. \quad (28)$$

In matrix notation (consistent with (24) *and* (25)*), the fuzzy dataset can be written as*

$$\tilde{\mathbf{y}} = \begin{bmatrix} 2.8 \\ 0.4 \\ \vdots \\ 6.1 \\ 2.7 \end{bmatrix}_{8\times 1} \quad \text{and} \quad \mathbf{G} = \begin{bmatrix} 1 & 1.0 & 2.0 \\ 0 & 0.3 & 0.8 \\ \vdots & \vdots & \vdots \\ 1 & 4.0 & 3.0 \\ 0 & 0.9 & 0.6 \end{bmatrix}_{8\times 3}.$$

Following the normal equations (26)*, the first step is to construct the matrices*

$$\mathbf{G}^{\mathsf{T}}\mathbf{G} = \begin{bmatrix} 4.00 & 10.00 & 9.00 \\ 10.00 & 32.59 & 26.35 \\ 9.00 & 26.35 & 26.12 \end{bmatrix}_{3\times 3} \quad \text{and} \quad \mathbf{G}^{\mathsf{T}}\tilde{\mathbf{y}} = \begin{bmatrix} 17.60 \\ 54.11 \\ 45.54 \end{bmatrix}_{3\times 1}.$$

The fuzzy least squares estimate of the real coefficients then follows by solving the system of normal equations (26)*. Substituting the matrices above yields the corresponding real coefficient vector*

$$\hat{\boldsymbol{\alpha}} = \begin{bmatrix} \hat{\alpha}_0 \\ \hat{\alpha}_1 \\ \hat{\alpha}_2 \end{bmatrix}_{3\times 1} \approx \begin{bmatrix} 0.867 \\ 1.227 \\ 0.207 \end{bmatrix}_{3\times 1}.$$

Based on (28)*, the corresponding predicted fuzzy responses are*

$$\hat{y}_1 \approx 2.508 + 0.534A, \quad \hat{y}_2 \approx 3.631 + 1.493A,$$
$$\hat{y}_3 \approx 5.065 + 1.007A, \quad \hat{y}_4 \approx 6.396 + 1.229A.$$

The fit quality can be quantified by Equations (13)*. In this case, we obtain*

$$\mathrm{MAE}_{\|\cdot\|_\Phi} = \frac{1}{4}\sum_{i=1}^{4} \|y_i \ominus \hat{y}_i\|_\Phi \approx 0.561,$$
$$\mathrm{RMSE}_{\|\cdot\|_\Phi} = \sqrt{\frac{1}{4}\sum_{i=1}^{4} \|y_i \ominus \hat{y}_i\|_\Phi^2} \approx 0.781.$$

In contrast, when the regression coefficients are treated as fuzzy numbers, their estimation must account for the fact that they belong to the field $\mathbb{R}_{\mathcal{F}(A)}$, and thus $\mathbb{R}^p_{\mathcal{F}(A)}$ forms a Hilbert space. This is discussed in the next subsection.

B. Fuzzy Multiple Linear Least Squares with Fuzzy Coefficients in $\mathbb{R}_{\mathcal{F}(A)}$

The fuzzy least squares method consists in determining $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \ldots, \alpha_n)^{\mathsf{T}} \in \mathbb{R}^{n+1}_{\mathcal{F}(A)}$ so that the norm of the fuzzy residual ($\|\mathbf{y} \ominus \mathbf{X} \odot \boldsymbol{\alpha}\|^2_\Phi$) is minimized. Equivalently, $\boldsymbol{\alpha}$ is obtained as the solution of the Problem (17), where the objective function $F : \mathbb{R}^{n+1}_{\mathcal{F}(A)} \to \mathbb{R}$ is defined over a fuzzy domain. Therefore, classical differential calculus cannot be applied directly to obtain the minimizer of F. Nevertheless, $\mathbb{R}^p_{\mathcal{F}(A)}$ endowed with the inner product introduced in Definition (1), is a Hilbert space [14], [18].

Because the fuzzy model (14) is linear in the fuzzy parameters and the column space of $\mathbf{X}$ is a linear subspace of $\mathbb{R}^p_{\mathcal{F}(A)}$, the fuzzy estimator $\hat{\boldsymbol{\alpha}}$ is such that the predicted fuzzy response $\hat{\mathbf{y}} = \mathbf{X} \odot \hat{\boldsymbol{\alpha}}$ corresponds to the orthogonal projection of $\mathbf{y}$ onto the subspace spanned by the columns of $\mathbf{X}$ (in complete analogy with the complex Hilbert space setting [23], due to the isometric isomorphism given in Theorem 2). Consequently, $\hat{\boldsymbol{\alpha}}$ is the solution of normal system

$$(\mathbf{X}^{\mathsf{H}} \odot \mathbf{X}) \odot \boldsymbol{\alpha} = \mathbf{X}^{\mathsf{H}} \odot \mathbf{y} \tag{29}$$

or

$$\hat{\boldsymbol{\alpha}} = (\mathbf{X}^{\mathsf{H}} \odot \mathbf{X})^{-1} \odot (\mathbf{X}^{\mathsf{H}} \odot \mathbf{y}), \tag{30}$$

whenever $\mathbf{X}^{\mathsf{H}} \odot \mathbf{X}$ is nonsingular, with H denoting the Hermitian (conjugate transpose) operator.

Example 2. *Using the same dataset as in Example 1, now we allow the coefficients to be fuzzy numbers, i.e., $\alpha_j \in \mathbb{R}_{\mathcal{F}(A)}$ for $j = 0, 1, 2$. The fuzzy multiple linear model takes the form*

$$y_i = \alpha_0 \oplus \alpha_1 \odot x_{i1} \oplus \alpha_2 \odot x_{i2}, \tag{31}$$

In matrix notation, we represent

$$\mathbf{y} = \begin{bmatrix} 2.8 + 0.4A \\ 3.7 + 1.3A \\ 5.0 + 0.8A \\ 6.1 + 2.7A \end{bmatrix}_{4\times 1}, \mathbf{X} = \begin{bmatrix} 1 & 1.0 + 0.3A & 2.0 + 0.8A \\ 1 & 2.0 + 1.2A & 1.5 + 0.1A \\ 1 & 3.0 + 0.5A & 2.5 + 1.9A \\ 1 & 4.0 + 0.9A & 3.0 + 0.6A \end{bmatrix}_{4\times 3}.$$

Applying the normal equations (29)*, we explicitly construct the matrices*

$$\mathbf{X}^{\mathsf{H}} \odot \mathbf{X} = \begin{bmatrix} 4.00 + 0.00A & 10.00 + 2.90A & 9.00 + 3.40A \\ 10.00 - 2.90A & 32.59 + 0.00A & 26.35 + 2.75A \\ 9.00 - 3.40A & 26.35 - 2.75A & 26.12 + 0.00A \end{bmatrix}_{3\times 3},$$

$$\mathbf{X}^{\mathsf{H}} \odot \mathbf{y} = \begin{bmatrix} 17.60 + 5.20A \\ 54.11 + 2.93A \\ 45.54 - 2.92A \end{bmatrix}_{3\times 1}.$$

Then, solving this system yields the fuzzy coefficient vector

$$\hat{\boldsymbol{\alpha}} = \begin{bmatrix} \hat{\alpha}_0 \\ \hat{\alpha}_1 \\ \hat{\alpha}_2 \end{bmatrix}_{3\times 1} \approx \begin{bmatrix} 2.250 - 1.086A \\ 1.233 + 0.460A \\ -0.182 + 0.221A \end{bmatrix}_{3\times 1}.$$

Each fuzzy coefficient $\hat{\alpha}_j = \alpha_j^{(r)} + \alpha_j^{(f)}A$, with $\alpha_j^{(r)}, \alpha_j^{(f)} \in \mathbb{R}$, admits a natural interpretation within the proposed framework. When the generating asymmetric fuzzy number A is centered at zero, it may be interpreted as a noise term. In this case, $\alpha_j^{(r)}$ represents the central value of the coefficient, while $\alpha_j^{(f)}$ acts as a scaling factor of the fuzzy variability around this central value. Using these coefficients and (31)*, the predicted values are*

$$\hat{y}_1 \approx 2.803 + 0.040A, \quad \hat{y}_2 \approx 3.868 + 1.627A,$$
$$\hat{y}_3 \approx 4.842 + 1.116A, \quad \hat{y}_4 \approx 6.087 + 2.417A.$$

Thus, we obtain the values

$$\mathrm{MAE}_{\|\cdot\|_\Phi} = \frac{1}{4}\sum_{i=1}^{4} \|y_i \ominus \hat{y}_i\|_\Phi \approx 0.340,$$
$$\mathrm{RMSE}_{\|\cdot\|_\Phi} = \sqrt{\frac{1}{4}\sum_{i=1}^{4} \|y_i \ominus \hat{y}_i\|_\Phi^2} \approx 0.343.$$

Comparing the two estimation scenarios, the fuzzy coefficient model achieves a lower $\mathrm{RMSE}_{\|\cdot\|_\Phi}$ of 0.343 than the real coefficient model $\mathrm{RMSE}_{\|\cdot\|_\Phi}$ of 0.781. Additionally, the $\mathrm{MAE}_{\|\cdot\|_\Phi}$ values follow the same pattern, decreasing from

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0.561 in the real coefficient case to 0.340 in the fuzzy coefficient case. Together, these results indicate an improved fit when the coefficients are treated as fuzzy numbers. This result is expected, since $\mathbb{R}^{n+1} \subset \mathbb{R}^{n+1}_{\mathcal{F}(A)}$.

In the following section, we illustrate both cases, real and fuzzy coefficients, through numerical examples and comparisons with the literature, highlighting how data can be represented and manipulated within $\mathbb{R}_{\mathcal{F}(A)}$. To convert data from the LR-fuzzy number representation $(m; l; u)$ to $\mathbb{R}_{\mathcal{F}(A)}$, we use the formula

$$m + \frac{l+u}{2}A, \tag{32}$$

where l and u denote the left and right spreads of the LR-fuzzy number, and $A = (m-l; m; m+u)_{LR} = (-0.99; 0; 1)_{LR}$ is a quasi-symmetric fuzzy number. This choice preserves the central tendency and maintains the uncertainty width.

V. Numerical Applications

We begin by examining the case in which the regression coefficients are real numbers.

A. Numerical Applications with Real Coefficients in $\mathbb{R}_{\mathcal{F}(A)}$

Example 3. *The case study from [12] examines employee engagement using fuzzy survey data from 25 employees of a Tax Administration agency in Iran. The goal was to build a fuzzy multiple linear regression model that captures the uncertainty in key individual factors related to engagement and job performance. Participants answered a 28 item questionnaire evaluated with triangular fuzzy numbers in $[0, 100]$. These items were grouped into seven dimensions, and for each dimension the fuzzy mean of its four items was computed. Although seven dimensions were originally measured, the analysis focused on the four most relevant ones, leading to the following fuzzy explanatory variables*

X_1 : *Job stress,* X_2 : *Delay frequency,*
X_3 : *Communication,* X_4 : *Performance.*

The fuzzy dependent variable Y represents each employee's assessed work quality. The fuzzy multiple linear model proposed in [12] is given by

$$Y = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 + \alpha_4 X_4, \tag{33}$$

and the intercept α_0 is fuzzy, while the coefficients α_j $(j = 1, 2, 3, 4)$ are real. In contrast, within the $\mathbb{R}_{\mathcal{F}(A)}$ framework, the fuzzy model is expressed as

$$Y_{\mathbb{R}_{\mathcal{F}(A)}} = \alpha_0 \oplus \alpha_1 X_1 \oplus \alpha_2 X_2 \oplus \alpha_3 X_3 \oplus \alpha_4 X_4, \tag{34}$$

with all coefficients α_j $(j = 0, 1, 2, 3, 4)$ treated as real values. From Equation (32), the resulting dataset consisting of 25 fuzzy input–output observations expressed in this format, is displayed in Table I.

Following the methodology of fuzzy least squares proposed in [12], the resulting fuzzy model is

$$\begin{aligned} Y = &(-131.94; 6.54; 6.42)_T \\ &- 0.013X_1 + 1.343X_2 + 0.607X_3 + 0.864X_4. \end{aligned} \tag{35}$$

Table I: Fuzzy input-output data in the employee engagement study. Original symmetric or asymmetric triangular fuzzy numbers and their corresponding representation in $\mathbb{R}_{\mathcal{F}(A)}$, with $A = (m-l; m; m+u)_T = (-0.99; 0; 1)_T$. The complete dataset (25 observations) is available in [12].

X_1	X_2	X_3	X_4	Y
$(87;8;8)_T$ $87+8A$	$(71;6;2)_T$ $71+4A$	$(86;4;5)_T$ $86+4.5A$	$(93;7;3)_T$ $93+5A$	$(95;5;3)_T$ $95+4A$
$(79;8;8)_T$ $79+8A$	$(80;6;3)_T$ $80+4.5A$	$(75;4;4)_T$ $75+4A$	$(67;2;3)_T$ $67+2.5A$	$(77;7;6)_T$ $77+6.5A$
$(24;8;8)_T$ $24+8A$	$(65;6;2)_T$ $65+4A$	$(71;2;4)_T$ $71+3A$	$(35;6;5)_T$ $35+5.5A$	$(26;3;6)_T$ $26+4.5A$
		⋮		
$(74;4;6)_T$ $74+5A$	$(61;2;5)_T$ $61+3.5A$	$(61;2;5)_T$ $61+3.5A$	$(82;5;3)_T$ $82+4A$	$(74;5;4)_T$ $74+4.5A$
$(63;3;2)_T$ $63+2.5A$	$(52;3;2)_T$ $52+2.5A$	$(58;2;3)_T$ $58+2.5A$	$(62;2;2)_T$ $62+2A$	$(22;3;3)_T$ $22+3A$

On the other hand, our fuzzy model in $\mathbb{R}_{\mathcal{F}(A)}$ using (26) yielded

$$Y_{\mathbb{R}_{\mathcal{F}(A)}} = -94.952 \oplus 0.189X_1 \oplus 1.029X_2 \oplus 0.272X_3 \oplus 0.752X_4. \tag{36}$$

*Table II presents the performance of our method and several methods found in the literature, based on the distance measure D_H introduced in [12]. As can be seen, our method achieved the lowest RMSE$_{D_H}$ of 12.228 among all compared approaches, indicating a significant improvement in predictive accuracy and a better fit to the data. Additionally, using the $\mathbb{R}_{\mathcal{F}(A)}$ norm, the **RMSE$_{\|\cdot\|_\Phi}$ was 13.105**, maintaining a low value compared to other methods. The corresponding **MAE$_{\|\cdot\|_\Phi}$ was 11.703**, further supporting the validity of the proposed formulation.*

In the following examples, we complement the RMSE analysis with similarity measures [13] to further assess the agreement between observed and predicted fuzzy responses.

Example 4. *Following the study of [13], we consider the fuzzy multiple linear model*

$$Y = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3, \tag{37}$$

where both the explanatory variables X_i $(i = 1, 2, 3)$ and the response Y are normal fuzzy numbers. The intercept α_0 is a normal fuzzy number, whereas the coefficients α_j $(j = 1, 2, 3)$ are real values to be estimated. In the $\mathbb{R}_{\mathcal{F}(A)}$ approach, however, the intercept is assumed to be real as well, leading to the fuzzy model

$$Y_{\mathbb{R}_{\mathcal{F}(A)}} = \alpha_0 \oplus \alpha_1 X_1 \oplus \alpha_2 X_2 \oplus \alpha_3 X_3. \tag{38}$$

In this example, we use quasi-normal fuzzy numbers (11) with $l = 0.99$ and $u = 1$, and express the fuzzy data as

$$B = m + \sigma A, \quad A = (m-l; m; m+l)_N = (-0.99; 0; 1)_N. \tag{39}$$

The experimental data and their representation in $\mathbb{R}_{\mathcal{F}(A)}$ are reported in Table III.

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Table II: Real coefficients of the model in $\mathbb{R}_{\mathcal{F}(A)}$ and RMSE based on the D_H distance for several fuzzy least squares models.

Method	$\hat{\alpha}_1$	$\hat{\alpha}_2$	$\hat{\alpha}_3$	$\hat{\alpha}_4$	$\hat{\alpha}_5$	**RMSE**$_{D_H}$
Chachi et al. [11]	-130.440	0.470	1.310	0.500	0.370	33.630
D'Urso et al. [9]	-115.030	0.238	1.308	0.568	0.824	26.800
Hesamian et al. [12]	$(-131.94;\, 6.54,\, 6.42)_T$	-0.013	1.343	0.607	0.864	25.980
Proposed	-94.952	0.189	1.029	0.272	0.752	**12.228**

Table III: Fuzzy input-output data from [13] and their corresponding representation in $\mathbb{R}_{\mathcal{F}(A)}$, with $A = (m-l; m; m+u)_N = (-0.99; 0; 1)_N$.

X_1	X_2	X_3	Y
$(3.00; 0.30)_N$ $3.00 + 0.30A$	$(3.20; 0.45)_N$ $3.20 + 0.45A$	$(3.10; 0.46)_N$ $3.10 + 0.46A$	$(30.80; 2.30)_N$ $30.80 + 2.30A$
$(2.20; 0.28)_N$ $2.20 + 0.28A$	$(2.80; 0.28)_N$ $2.80 + 0.28A$	$(2.50; 0.30)_N$ $2.50 + 0.30A$	$(26.60; 2.00)_N$ $26.60 + 2.00A$
$(4.50; 0.30)_N$ $4.50 + 0.30A$	$(1.80; 0.15)_N$ $1.80 + 0.15A$	$(3.20; 0.23)_N$ $3.20 + 0.23A$	$(32.80; 4.90)_N$ $32.80 + 4.90A$
$(4.10; 0.60)_N$ $4.10 + 0.60A$	$(2.90; 0.43)_N$ $2.90 + 0.43A$	$(3.50; 0.50)_N$ $3.50 + 0.50A$	$(32.50; 3.60)_N$ $32.50 + 3.60A$
$(4.90; 0.63)_N$ $4.90 + 0.63A$	$(3.40; 0.15)_N$ $3.40 + 0.15A$	$(4.20; 0.40)_N$ $4.20 + 0.40A$	$(36.40; 2.70)_N$ $36.40 + 2.70A$
$(2.40; 0.23)_N$ $2.40 + 0.23A$	$(4.00; 0.50)_N$ $4.00 + 0.50A$	$(3.20; 0.37)_N$ $3.20 + 0.37A$	$(26.10; 2.00)_N$ $26.10 + 2.00A$
$(3.80; 0.38)_N$ $3.80 + 0.38A$	$(2.10; 0.20)_N$ $2.10 + 0.20A$	$(2.90; 0.26)_N$ $2.90 + 0.26A$	$(25.10; 1.90)_N$ $25.10 + 1.90A$
$(2.20; 0.15)_N$ $2.20 + 0.15A$	$(3.00; 0.23)_N$ $3.00 + 0.23A$	$(2.60; 0.18)_N$ $2.60 + 0.18A$	$(22.00; 1.70)_N$ $22.00 + 1.70A$
$(4.60; 0.45)_N$ $4.60 + 0.45A$	$(1.40; 0.13)_N$ $1.40 + 0.13A$	$(3.00; 0.29)_N$ $3.00 + 0.29A$	$(26.90; 4.10)_N$ $26.90 + 4.10A$
$(3.40; 0.35)_N$ $3.40 + 0.35A$	$(3.40; 0.50)_N$ $3.40 + 0.50A$	$(3.70; 0.36)_N$ $3.70 + 0.36A$	$(26.80; 3.40)_N$ $26.80 + 3.40A$

Using two distance measures between LR-type fuzzy numbers, [13] proposed a fuzzy multiple linear least squares estimation method. Their LR-metric–based minimization approach leads to the following fuzzy multiple linear regression models

$$Y_{D_{YK}} = (6.16; 0.39)_N + 7.61X_1 + 6.47X_2 - 7.02X_3, \quad (40)$$
$$Y_{D_{DK}} = (6.75; 0.47)_N + 7.20X_1 + 5.99X_2 - 6.33X_3. \quad (41)$$

On the other hand, the fuzzy model obtained from our method by solving (26) *is given by*

$$Y_{\mathbb{R}_{\mathcal{F}(A)}} = 5.084 \oplus 7.984X_1 \oplus 6.852X_2 \ominus 7.428X_3. \quad (42)$$

As shown in Table IV, which also presents the fuzzy outputs, our fuzzy multiple linear regression model ($Y_{\mathbb{R}_{\mathcal{F}(A)}}$) achieves a slightly lower RMSE than the models reported in [13]. Furthermore, by applying the $\|\cdot\|_\Phi$ norm in $\mathbb{R}_{\mathcal{F}(A)}$, the ***RMSE**$_{\|\cdot\|_\Phi}$ **is reduced to 2.612**. The associated **MAE**$_{\|\cdot\|_\Phi}$ **is 2.316**. This enhancement highlights an important advantage of working directly within the $\mathbb{R}_{\mathcal{F}(A)}$ framework.*

Regarding similarity, Table V summarizes the similarity measures for the 3 fuzzy regression approaches, providing a direct comparison of their performance.

The similarity results reinforce the conclusions drawn from the RMSE analysis: on average, the $\mathbb{R}_{\mathcal{F}(A)}$ model exhibits the highest level of agreement with the observed fuzzy data.

Table IV: Fuzzy output and RMSE comparison: our method with $A = (-0.99; 0; 1)_N$ vs the method from [13].

Y	$\hat{Y}_{D_{YK}}$	$\hat{Y}_{D_{DK}}$	$\hat{Y}_{\mathbb{R}_{\mathcal{F}(A)}}$
$(30.80; 2.30)_N$	$(27.93; 3.06)_N$	$(27.90; 3.05)_N$	$27.93 + 2.80A$
$(26.60; 2.00)_N$	$(23.47; 2.23)_N$	$(23.54; 2.26)_N$	$23.26 + 1.92A$
$(32.80; 4.90)_N$	$(29.59; 2.03)_N$	$(29.68; 2.07)_N$	$29.57 + 1.71A$
$(32.50; 3.60)_N$	$(31.55; 4.23)_N$	$(31.49; 4.20)_N$	$31.69 + 4.02A$
$(36.40; 2.70)_N$	$(35.96; 3.35)_N$	$(35.81; 3.37)_N$	$36.30 + 3.08A$
$(26.10; 2.00)_N$	$(27.84; 2.78)_N$	$(27.73; 2.78)_N$	$27.88 + 2.47A$
$(25.10; 1.90)_N$	$(28.31; 2.75)_N$	$(28.33; 2.76)_N$	$28.27 + 2.47A$
$(22.00; 1.70)_N$	$(24.06; 1.76)_N$	$(24.10; 1.79)_N$	$23.89 + 1.43A$
$(26.90; 4.10)_N$	$(29.16; 2.62)_N$	$(29.27; 2.65)_N$	$29.12 + 2.32A$
$(26.80; 3.40)_N$	$(28.06; 3.76)_N$	$(28.18; 3.71)_N$	$28.04 + 3.54A$
RMSE$_{D_{YK}}$	5.900	5.830	**5.632**
RMSE$_{D_{DK}}$	3.980	3.930	**3.811**

Table V: Similarity values S'_4, and S^*_4 for $\hat{Y}_{D_{YK}}$, $\hat{Y}_{D_{DK}}$ and $\hat{Y}_{\mathbb{R}_{\mathcal{F}(A)}}$.

	$\hat{Y}_{D_{YK}}$		$\hat{Y}_{D_{DK}}$		$\hat{Y}_{\mathbb{R}_{\mathcal{F}(A)}}$	
	S'_4	S^*_4	S'_4	S^*_4	S'_4	S^*_4
1	0.300	0.448	0.280	0.440	0.403	0.575
2	0.173	0.295	0.180	0.310	0.246	0.395
3	0.302	0.464	0.315	0.480	0.310	0.473
4	0.756	0.861	0.740	0.850	0.843	0.914
5	0.790	0.882	0.766	0.870	0.880	0.936
6	0.440	0.606	0.459	0.630	0.523	0.686
7	0.200	0.329	0.195	0.330	0.302	0.464
8	0.250	0.399	0.246	0.390	0.372	0.542
9	0.460	0.628	0.440	0.620	0.501	0.668
10	0.660	0.804	0.640	0.780	0.747	0.855
Mean	0.433	0.572	0.426	0.570	**0.513**	**0.651**

These results from both examples clearly indicate that the proposed method not only reduces estimation errors but also provides a computationally efficient and robust structure within the $\mathbb{R}_{\mathcal{F}(A)}$ framework. The computational advantage arises from performing all algebraic operations through the isomorphism Φ, enabling the fuzzy least squares problem to be solved directly using standard matrix operations. Previous fuzzy least squares formulations, such as those in [12], [13], often rely on α-cut decomposition or iterative optimization procedures, which are naturally required within their respective frameworks.

In the following subsection, we consider the case where the coefficients themselves are fuzzy.

B. Numerical Applications with Fuzzy Coefficients in $\mathbb{R}_{\mathcal{F}(A)}$

Example 5. *Diamond et al. [5] presented an application involving a fuzzy multiple linear regression model with data represented by symmetric or asymmetric triangular fuzzy numbers of the form $(m; l; u)_T$, where m denotes the central value and l, u are the left and right spreads, respectively. He related hospital admission patterns (Y) to health service availability indices (X_1) and indigence (X_2) for 10 communities in the hospital's service area.*

From Equation (32), the data is converted to the $\mathbb{R}_{\mathcal{F}(A)}$ format, resulting in the dataset presented in Table VI.

Table VI: Fuzzy input-fuzzy output data from Diamond's hospital admission study, represented in triangular form $(m; l; u)_T$ and their corresponding representations in $\mathbb{R}_{\mathcal{F}(A)}$, with $A = (m-l; m; m+u)_T = (-0.99; 0; 1)_T$.

X_1	X_2	Y
$(6.00; 0.30; 0.90)_T$ $6.00 + 0.60A$	$(6.30; 0.90; 0.90)_T$ $6.30 + 0.90A$	$(61.60; 6.20; 3.10)_T$ $61.60 + 4.65A$
$(4.40; 0.40; 0.70)_T$ $4.40 + 0.55A$	$(5.50; 0.80; 0.30)_T$ $5.50 + 0.55A$	$(53.20; 2.70; 5.30)_T$ $53.20 + 4.00A$
$(9.10; 0.50; 0.70)_T$ $9.10 + 0.60A$	$(3.60; 0.20; 0.40)_T$ $3.60 + 0.30A$	$(65.50; 9.80; 9.80)_T$ $65.50 + 9.80A$
$(8.10; 1.20; 1.20)_T$ $8.10 + 1.20A$	$(5.80; 0.80; 0.80)_T$ $5.80 + 0.80A$	$(64.30; 9.30; 9.80)_T$ $64.30 + 9.55A$
$(9.70; 1.00; 1.50)_T$ $9.70 + 1.25A$	$(6.80; 0.30; 0.30)_T$ $6.80 + 0.30A$	$(72.70; 3.60; 7.30)_T$ $72.70 + 5.45A$
$(4.80; 0.20; 0.70)_T$ $4.80 + 0.45A$	$(7.90; 1.80; 0.30)_T$ $7.90 + 1.05A$	$(52.20; 2.60; 5.20)_T$ $52.20 + 3.90A$
$(7.60; 0.40; 1.10)_T$ $7.60 + 0.75A$	$(4.20; 0.20; 0.60)_T$ $4.20 + 0.40A$	$(50.20; 2.50; 5.00)_T$ $50.20 + 3.75A$
$(4.40; 0.40; 0.40)_T$ $4.40 + 0.40A$	$(6.00; 0.60; 0.30)_T$ $6.00 + 0.45A$	$(44.00; 2.20; 4.40)_T$ $44.00 + 3.30A$
$(9.10; 0.90; 0.90)_T$ $9.10 + 0.90A$	$(2.80; 0.10; 0.40)_T$ $2.80 + 0.25A$	$(53.80; 8.10; 8.10)_T$ $53.80 + 8.10A$
$(6.70; 0.70; 0.70)_T$ $6.70 + 0.70A$	$(6.70; 1.00; 1.00)_T$ $6.70 + 1.00A$	$(53.50; 8.10; 5.40)_T$ $53.50 + 6.75A$

Following the general fuzzy multiple linear regression model

$$Y = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2, \tag{43}$$

the estimation obtained by Diamond et al. [5] is given by

$$Y_D = (13.96; 0.90; 1.21)_T + 3.97X_1 + 2.78X_2. \tag{44}$$

In this fuzzy model, the intercept α_0 is a triangular fuzzy number, whereas the coefficients α_1 and α_2 are real. This choice is not arbitrary but follows directly from the algebraic structure underlying the fuzzy least squares method developed in [5]. The procedure is formulated in the Hilbert space $L^2(S^{d-1} \times [0,1])$, where the class of LR-type fuzzy numbers forms only a convex cone rather than a complete linear space. As a consequence, scalar multiplication is well defined only for nonnegative real scalars, and the product of two fuzzy numbers is not closed within the LR-class. Therefore, while the intercept α_0 may be fuzzy, the regression coefficients α_1 and α_2 must be real to ensure that the terms $\alpha_1 X$ and $\alpha_2 Z$ in (43) remain LR-type fuzzy numbers. Allowing fuzzy coefficients would violate the cone structure of the model space and prevent the application of the Hilbert projection theorem [23] that guarantees existence and uniqueness of the fuzzy least squares solution. In fact, this limitation constitutes one of the main motivations of the present work, namely, the search for a class of fuzzy numbers, which constitutes a space of Hilbert in which all regression coefficients can consistently be modeled as fuzzy quantities.

Following up, consider the fuzzy model in the $\mathbb{R}_{\mathcal{F}(A)}$ space

$$Y_{\mathbb{R}_{\mathcal{F}(A)}} = \alpha_0 \oplus \alpha_1 \odot X_1 \oplus \alpha_2 \odot X_2, \tag{45}$$

where all parameters α_0, α_1 and α_2 are in $\mathbb{R}_{\mathcal{F}(A)}$, that is, are fuzzy numbers. Alternatively to Model (44), in Equation (45) it is not necessary to restrict the regression coefficients to real values, since all arithmetic operations are well defined for fuzzy numbers in $\mathbb{R}_{\mathcal{F}(A)}$, according to Theorem 2. Moreover, the isometric isomorphism Φ ensures that the norm and inner product on $\mathbb{R}_{\mathcal{F}(A)}$ inherit the standard Hilbert space properties, enabling the direct application of fuzzy least squares theory to models with fuzzy-valued parameters, as developed in Section IV.

Based on the fuzzy data in Table VI, we estimated the following fuzzy model using Equation (29)

$$\begin{aligned} Y_{\mathbb{R}_{\mathcal{F}(A)}} &= (13.333 + 3.779A) \\ &\oplus\ (4.043 + 0.290A) \odot X_1 \oplus\ (2.738 - 0.812A) \odot X_2. \end{aligned} \tag{46}$$

Table VII presents the observed values, the estimates obtained from Equation (44) (Diamond et al [5]), and those obtained via Equation (45), that is, from $\mathbb{R}_{\mathcal{F}(A)}$ approach. Residuals were analyzed based on Diamond's distance, which was employed as the metric for quantifying deviations between fuzzy observations and model predictions (the same distance D_{DK} as in Example 4).

Table VII: Comparison of observed fuzzy outputs (Y), estimates obtained from Diamond et al. model $(Y_{D_{DK}})$, and predictions derived using the $\mathbb{R}_{\mathcal{F}(A)}$ approach $(Y_{\mathbb{R}_{\mathcal{F}(A)}})$, along with the corresponding RMSE values, and $A = (m-l; m; m+u)_T = (-0.99; 0; 1)_T$.

Y	$Y_{D_{DK}}$	$Y_{\mathbb{R}_{\mathcal{F}(A)}}$
$(61.60; 6.20; 3.10)_T$	$(55.29; 4.59; 7.29)_T$	$55.41 + 5.29A$
$(53.20; 2.70; 5.30)_T$	$(46.72; 4.71; 4.82)_T$	$46.47 + 4.31A$
$(65.50; 9.80; 9.80)_T$	$(60.09; 3.72; 5.10)_T$	$60.06 + 6.74A$
$(64.30; 9.30; 9.80)_T$	$(62.24; 7.33; 6.53)_T$	$62.27 + 8.45A$
$(72.70; 3.60; 7.30)_T$	$(69.98; 7.37; 9.39)_T$	$71.06 + 6.94A$
$(52.20; 2.60; 5.20)_T$	$(54.98; 5.03; 4.54)_T$	$55.10 + 3.44A$
$(50.20; 2.50; 5.00)_T$	$(55.81; 3.04; 7.25)_T$	$55.67 + 6.69A$
$(44.00; 2.20; 4.40)_T$	$(48.11; 3.64; 5.10)_T$	$47.80 + 3.03A$
$(53.80; 8.10; 8.10)_T$	$(57.87; 4.75; 5.06)_T$	$57.74 + 8.46A$
$(53.50; 8.10; 5.40)_T$	$(59.19; 6.46; 4.54)_T$	$59.38 + 5.84A$
RMSE$_{D_{DK}}$	8.380	**8.056**

The results show that the $\mathbb{R}_{\mathcal{F}(A)}$ model achieves a lower RMSE$_{D_{DK}}$ of 8.056, compared to 8.380 from the baseline method, confirming the gain in accuracy obtained with the proposed formulation. Furthermore, when considering the

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*norm in $\mathbb{R}_{\mathcal{F}(A)}$, the **RMSE**$_{\|\cdot\|_\Phi}$ **further decreases to 4.962**, while the **MAE**$_{\|\cdot\|_\Phi}$ **is 4.660**. This reinforces the robustness of the proposed approach.*

To assess the agreement between observed and predicted fuzzy outputs, we employ the similarity measure from [13]. The corresponding results are presented in Table VIII.

Table VIII: Similarity values between observed fuzzy outputs Y and estimates via Diamond et al. and $\mathbb{R}_{\mathcal{F}(A)}$ approaches.

	$S_4(Y, Y_{D_{DK}})$	$S_4(Y, Y_{\mathbb{R}_{\mathcal{F}(A)}})$
1	0.522	0.664
2	0.702	0.694
3	0.340	0.513
4	0.598	0.781
5	0.558	0.698
6	0.716	0.749
7	0.695	0.531
8	0.759	0.763
9	0.518	0.873
10	0.712	0.693
Mean	0.612	**0.696**

According to Table VIII, the $\mathbb{R}_{\mathcal{F}(A)}$ formulation consistently achieves higher similarity values, with a mean S_4 of 0.696, compared to 0.612 for the Diamond et al. fuzzy model. This improvement corroborates the RMSE analysis and further suggests that the $\mathbb{R}_{\mathcal{F}(A)}$ method more accurately captures both the structure and the uncertainty of the observed fuzzy data.

VI. Conclusion

This article addresses the fuzzy least squares method for fuzzy multiple linear regression models. Unlike traditional formulations based on the extension principle or LR-type fuzzy numbers, the fuzzy data considered here can be written in the form $r + qA$, which defines the class of A-linearly correlated fuzzy numbers, denoted by $\mathbb{R}_{\mathcal{F}(A)}$. In this representation, all data inherit the shape of the reference fuzzy number A, which may be triangular, trapezoidal, normal, sigmoidal, and so forth. A key feature of this class is its suitability for fuzzy least squares analysis. A key feature that makes this class particularly suitable for fuzzy least squares analysis is that, under the mild assumption that A is asymmetric (quasi-symmetric, for example), $\mathbb{R}_{\mathcal{F}(A)}$ becomes a Banach/Hilbert space, hence a normed space endowed with an inner product. This structural property follows from the fact that $\mathbb{R}_{\mathcal{F}(A)}$ is isomorphic to both the Euclidean plane $\mathbb{R}^2$ and the field of complex numbers $\mathbb{C}$ [17], [18].

Since $\mathbb{R}_{\mathcal{F}(A)}$ is a field, and therefore a vector space over itself, the full framework of differential and integral calculus for complex numbers extends naturally to $\mathbb{R}_{\mathcal{F}(A)}$. Owing to isomorphism, the least squares methodology in $\mathbb{R}_{\mathcal{F}(A)}$ (or in $\mathbb{R}^p_{\mathcal{F}(A)}$) follows the same script as in $\mathbb{R}^2$ or in the complex space $\mathbb{C}$ ($\mathbb{C}^p$). Accordingly, for Equation (18), whose coefficients are real numbers, we apply the standard maxima–minima theory in $\mathbb{R}^p$ to obtain the solution. Instead, for fuzzy coefficients in $\mathbb{R}_{\mathcal{F}(A)}$ (Equation (17)) the best approximation is rely via orthogonal projections. In both settings, fully analogous to the classical (real) case, the least squares solutions are determined by the corresponding normal equations, (26) and (29), respectively.

A computational advantage of the proposed method follows directly from the fact that the estimator is obtained by solving the normal equations, a linear system handled entirely through standard matrix operations available in any numerical linear algebra routine. Because $\mathbb{R}_{\mathcal{F}(A)}$ is isometrically isomorphic to $\mathbb{C}$, all algebraic manipulations involving fuzzy coefficients reduce to those for complex numbers, eliminating the need for α-level decompositions or the iterative update schemes typically required in traditional fuzzy least squares methods [5], [12], [13].

From a predictive performance perspective, we use the numerical examples presented in Section V to compare our methodology with well-established approaches in the literature ([5], [12], [13]), using the error measures originally adopted in those works. According to Tables II, IV, and VII, the proposed estimator achieves lower RMSE values, even when evaluated using the error measures originally adopted in those works. The improvement is even more pronounced under the $\mathbb{R}_{\mathcal{F}(A)}$ norm $\|\cdot\|_\Phi$. Moreover, the corresponding MAE values under the $\mathbb{R}_{\mathcal{F}(A)}$ norm remain close to the RMSE values, indicating a stable residual distribution without significant deviations. This trend is further corroborated by the similarity measures presented in Examples 4 and 5 (Tables V and VIII), where predictions based on $\mathbb{R}_{\mathcal{F}(A)}$ exhibit higher average similarities with the observed fuzzy outputs.

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